Unit-5 (Partial diff eq^n)

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$\frac{\delta z}{\delta x} = p$, $\frac{\delta z}{\delta y} = q$, $\frac{\delta^2 z}{\delta x^2} = r$, $\frac{\delta^2 z}{\delta x \delta y} = s$, $\frac{\delta^2 z}{\delta y \delta z} = t$

★ **Method of seperation of variable:**

→ We assume the sol^n to be the product of two different f^n's, each of which involves different variables.

$$\cancel{\frac{\delta u}{\delta x} = 2\frac{\delta u}{\delta t} + u}$$

Q- $\frac{\delta u}{\delta x} = 2\frac{\delta u}{\delta t} + u$, given $u(x,0) = 6e^{-3x}$.

→ Let, $u = XT$ $\qquad \frac{\delta u}{\delta t} = XT'$

$\frac{\delta u}{\delta x} = X'T$

→ $X'T = 2XT' + XT = X(2T' + T)$

$$\frac{X'}{X} = \frac{2T'+T}{T} = k$$

$\frac{X'}{X} = k$

$\int \frac{1}{X} dX = \int k\, dx$

→ $\ln X = kx + \ln c_1$

→ $\boxed{X = c_1 e^{kx}}$

$\frac{2T'+T}{T} = k$

$\frac{2T'}{T} = k-1$

$\frac{2}{T} dT = (k-1)\, dt$

$2 \ln T = (k-1)t + \ln c_2$

→ $\boxed{T = c_2 e^{\frac{(k-1)t}{2}}}$

$\Rightarrow \quad u(x,t) = C_1 C_2 \, e^{kx + \frac{(k-1)}{2}t}$

at $t=0$,

$u(x,0) = 6e^{-3x}$

$\rightarrow \quad C_1 C_2 \, e^{kx} = 6e^{-3x}$

$\rightarrow \quad [C_1 C_2 = 6], \quad [k = -3]$

$\Rightarrow \quad \boxed{u(x,t) = 6e^{-(3x+2t)}}$

Q - Solve, $\dfrac{\delta^2 z}{\delta x^2} - 2\dfrac{\delta z}{\delta x} + \dfrac{\delta z}{\delta y} = 0$

$\rightarrow \quad z = XY$

$\dfrac{\delta^2 z}{\delta x^2} = X''Y, \quad \dfrac{\delta z}{\delta x} = X'Y, \quad \dfrac{\delta z}{\delta y} = XY'$

$\rightarrow \quad X''Y - 2X'Y + XY' = 0$

$\rightarrow \quad (X'' - 2X')Y = -XY'$

$\rightarrow \quad \left(\dfrac{X''}{X} - \dfrac{2X'}{X}\right) = \dfrac{-Y'}{Y} = k$

$\dfrac{X''}{X} - \dfrac{2X'}{X} = k$

$X'' - 2X' - kX = 0 \quad \Rightarrow \quad \boxed{X = C_1 e^{(1+\sqrt{k+1})x} + C_2 e^{(1-\sqrt{k+1})x}}$

$m^2 - 2m - k = 0$

$m = \dfrac{2 \pm \sqrt{4+4k}}{2}$

$\rightarrow \quad \boxed{m = 1 \pm \sqrt{1+k}}$

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$$-\frac{Y'}{Y} = k$$

$$\rightarrow \ln Y = -ky + \ln c_3$$

$$\rightarrow \boxed{Y = c_3 e^{-ky}}$$

$$\left[Z = XY = c_1 c_3 e^{(1+\sqrt{k+1})x - ky} + c_2 c_3 e^{(1-\sqrt{k+1})x - ky} \right]$$

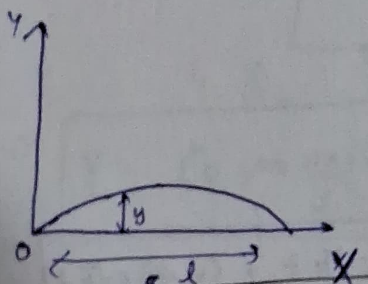
$$\rightarrow \boxed{Z = A e^{(1+\sqrt{k+1})x - ky} + B e^{(1-\sqrt{k+1})x - ky}}$$

* Note → Vibration of a stretched string (1-D wave eqⁿ) ~~as a f~~ We shall find displacement 'y' as a fⁿ of distance 'x' & time 't'.

$$\rightarrow \boxed{\frac{\delta^2 y}{\delta t^2} = c^2 \frac{\delta^2 y}{\delta x^2}}$$

Boundary:

$$\left. \begin{matrix} x = 0 \\ x = l \end{matrix} \right\} y = 0$$



If a string is made to vibrate by pulling it to a curve $y = f(x)$ & then releasing it, the initial conditions are

$$\left\{ \begin{matrix} y = f(x), & t = 0 \\ \frac{\delta y}{\delta t} = 0, & t = 0 \end{matrix} \right\}$$

Solⁿ of wave eqⁿ!

$$\frac{\delta^2 y}{\delta t^2} = c^2 \frac{\delta^2 y}{\delta x^2}$$

$$Y = XT$$

$$\rightarrow XT'' = c^2 X'' T$$

$$\rightarrow \left[\frac{X''}{X} c^2 = \frac{T''}{c^2 T} = k \right]$$

$$\frac{X''}{X} = k$$

$\rightarrow [m = \pm\sqrt{k}]$

$$\Rightarrow \boxed{X = C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x}}$$

$$\frac{T''}{c^2 T} = k$$

$$\frac{T''}{T} = c^2 k$$

$[m = \pm c\sqrt{k}]$

$$\Rightarrow \boxed{T = C_3 e^{c\sqrt{k}t} + C_4 e^{-c\sqrt{k}t}}$$

$$\Rightarrow y = XT$$

$$\rightarrow \boxed{y = \left(C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x}\right)\left(C_3 e^{c\sqrt{k}t} + C_4 e^{-c\sqrt{k}t}\right)} \rightarrow k > 0$$

For, $k < 0$

$$\frac{X''}{X} = k$$

$[m = \pm\sqrt{k}\,i]$

$$\Rightarrow \boxed{X = C_1 \cos\sqrt{k}x + C_2 \sin\sqrt{k}x}$$

$m = \pm c\sqrt{k}\,i$

$$\Rightarrow \boxed{T = C_3 \cos c\sqrt{k}t + C_4 \sin c\sqrt{k}t}$$

$$\Rightarrow y = XT.$$

$$\rightarrow \boxed{y = \left(C_1 \cos\sqrt{k}x + C_2 \sin\sqrt{k}x\right)\left(C_3 \cos c\sqrt{k}t + C_4 \sin\sqrt{k}t\right)}$$

Out of these solⁿs, we have to chose the solⁿ similar to the physical nature of problem. Since we are dealing with vibration of string, y must be a periodic fⁿ, thus the solⁿ of $k < 0$ is best solⁿ.

$$\Rightarrow \left\{ y = \left(C_1 \cos\sqrt{k}x + C_2 \sin\sqrt{k}x\right)\left(C_3 \cos c\sqrt{k}t + C_4 \sin\sqrt{k}t\right) \right\}$$

Initial cond^n, $x=0,\ y=0$

$\rightarrow\ 0 = C_1\left(C_3 \cos c\sqrt{k}t + C_4 \sin c\sqrt{k}t\right)$

$$\boxed{C_1 = 0}$$

$x = l,\ y = 0$,

$0 = \left(C_2 \sin\sqrt{k}l\right)\left(C_3 \cos c\sqrt{k}t + C_4 \sin c\sqrt{k}t\right)$

$\rightarrow\ C_2 \sin\sqrt{k}l = 0$

$\rightarrow\ \sqrt{k}l = n\pi$

$\rightarrow\ \boxed{k = \left(\frac{n\pi}{l}\right)^2}$

$$\Rightarrow\ \boxed{y = C_2 \sin\frac{n\pi x}{l}\left(C_3 \cos\frac{cn\pi}{l}t + C_4 \sin\frac{cn\pi}{l}t\right)}$$

$$\Downarrow$$

$$\boxed{y = \sin\frac{n\pi x}{l}\left(A_n \cos\frac{cn\pi}{l}t + B_n \sin\frac{cn\pi}{l}t\right)}$$

Adding up sol^{n}s for different values of n,

$$\Rightarrow\ \left[y = \sum_{n=1}^{\infty}\left(a_n \cos\frac{n\pi c}{l}t + b_n \sin\frac{n\pi c}{l}t\right)\sin\frac{n\pi}{l}x\right]$$

$t=0,\ y = f(x)$

$$\rightarrow\ \boxed{f(x) = \sum_{n=1}^{\infty} a_n \sin\frac{n\pi x}{l}} \longrightarrow \left\{a_n = \frac{2}{l}\int_0^l f(x)\sin\frac{n\pi x}{l}\,dx\right\}$$

$$\rightarrow \frac{\delta y}{\delta t} = \sum_{n=1}^{\infty} \left(\left(-a_n \sin \frac{n\pi c}{l} t\right)\left(\frac{l}{c\pi}\right) + b_n \left(\frac{l}{cn}\right) \cos \frac{n\pi c}{l} t \right) \sin \frac{n\pi x}{l}$$

$t=0, \ \frac{\delta y}{\delta t} = 0$

$$\rightarrow \sum_{n=1}^{\infty} b_n \left(\frac{l}{c\pi}\right) \sin \frac{n\pi x}{l} = 0$$

$$\Rightarrow \boxed{b_n = 0}$$

$$\Rightarrow \boxed{y = \sum_{n=1}^{\infty} a_n \cos \frac{n\pi c}{l} t \ \sin \frac{n\pi x}{l}}$$

Q- Solve : $\frac{\delta^2 y}{\delta t^2} = a^2 \frac{\delta y^2}{\delta x^2}$ for a string vibration of length 'l', fixed at both ends subjected to boundary $y(0,t) = 0$, $y(l,t) = 0$ & initial conditions,

$y = y_0 \sin \frac{\pi x}{l}$, $\left(\frac{\delta y}{\delta t} = 0, t=0\right)$.

$$\rightarrow y = (C_1 \cos \sqrt{k} x + C_2 \sin \sqrt{k} x)(C_3 \cos(a\sqrt{k} t) + C_4 \sin a\sqrt{k} t)$$

$y(0,t) = 0$

$$\rightarrow \boxed{C_1 = 0}$$

$y(l,t) = 0,$

$$\boxed{\sqrt{k} = \frac{n\pi}{l}}$$

$$\Rightarrow \left[y = \sin\frac{n\pi x}{l}\left(a_n \cos\frac{an\pi}{l} t + b_n \frac{an\pi}{l} t\right)\right]$$

$t = 0, \quad y = y_0 \sin\frac{\pi x}{l}$

$$\rightarrow \quad y_0 \sin\frac{\pi x}{l} = \sin\frac{n\pi x}{l}(a_n)$$

$$\Rightarrow \boxed{a_n = y_0}, \quad \boxed{n = 1}$$

$$\Rightarrow \quad y = \sin\frac{\pi x}{l}\left[y_0 \cos\frac{a\pi}{l} t + b_n \sin\left(\frac{a\pi}{l} t\right)\right]$$

$$\frac{\delta y}{\delta t} = \sin\frac{\pi x}{l}\left[\frac{y_0 a\pi}{l}\left(-\sin\frac{a\pi}{l} t\right) + b_n \frac{a\pi}{l}\cos\left(\frac{a\pi}{l} t\right)\right]$$

$t = 0,$

$$0 = \sin\frac{\pi x}{l}\left(b_n \frac{a\pi}{l}\right)$$

$$\rightarrow \boxed{b_n = 0}$$

$$\Rightarrow \quad y = \sin\frac{\pi x}{l}\left[y_0 \cos\frac{a\pi}{l} t\right]$$

$$\rightarrow \boxed{y = y_0 \sin\left(\frac{\pi x}{l}\right)\cos\left(\frac{a\pi}{l} t\right)}$$

Q- A string with fixed ~~n~~ n-end points $x=0$, $x=l$ is initially at rest. It will be a mistake if it is set vibrating by each of its pts. with velocity $\mu x(l-x)$. Find displacement of string from end 'x' at any pt. of time 't'.

$\rightarrow \quad y(0,t)=0 \quad , \quad y(l,t)=0$

$$y(x,t) = \sum_{n=1}^{\infty} \left(a_n \cos\frac{n\pi c t}{l} + b_n \sin\frac{n\pi c t}{l} \right) \sin\frac{n\pi x}{l}$$

$y(x,0)=0$

$$\left[0 = \sum_{n=1}^{\infty} a_n \sin\frac{n\pi x}{l} \right]$$

$$\Rightarrow \quad y(x,t) = \sum_{n=1}^{\infty} b_n \sin\frac{n\pi c t}{l} \sin\frac{n\pi x}{l}$$

$$\rightarrow \quad \frac{\delta y}{\delta t} = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \cos\frac{n\pi c t}{l} \sin\frac{n\pi x}{l}$$

at $t=0$,

$$\left[\mu x(l-x) = \sum_{n=1}^{\infty} b_n \left(\frac{n\pi c}{l}\right) \sin\frac{n\pi x}{l} \right] \rightarrow \text{half-wave sine series}$$

$$\Rightarrow \quad b_n\left(\frac{n\pi c}{l}\right) = \frac{2}{l}\int_0^l \mu x(l-x) \sin\frac{n\pi x}{l}\, dx$$

$$\rightarrow \quad b_n(n\pi c) = 2\mu \int_0^l (lx - x^2) \sin\frac{n\pi x}{l}\, dx$$

$$\rightarrow \quad b_n(n\pi c) = 2\mu \left[\left(lx \cdot -\cos\left(\frac{n\pi x}{l}\right)\left(\frac{l}{n\pi}\right) \right)_0^l - \left(l \cdot -\sin\frac{n\pi x}{l} \left(\frac{l^2}{n^2\pi^2}\right) \right)_0^l \right.$$
$$\left. + \left(x^2 \cdot +\cos\left(\frac{n\pi x}{l}\right)\left(\frac{l}{n\pi}\right) \right)_0^l + \left(2x \cdot -\sin\frac{n\pi x}{l} \left(\frac{l}{n\pi}\right)^2 \right)_0^l - \left(\left(2 \cos\left(\frac{n\pi x}{l}\right)\left(\frac{l}{n\pi}\right)^3 \right)_0^l \right) \right]$$

$\rightarrow b_n(n\pi c) = 24\left[-\frac{(-1)^n \cancel{l^3}}{n\pi} + \frac{(-1)^n l^3}{\cancel{n\pi}} - \left(\frac{2l^3(-1)^n}{(n\pi)^3} - \frac{2l^3}{(n\pi)^3}\right)\right]$

$$\rightarrow \boxed{b_n(n\pi c) = \frac{44 l^3}{(n\pi)^3}\left[1-(-1)^n\right]}$$

~~Q- Two pts l-apart are set~~

Q- A string is stretched & fastened across 2-pts. 'l' apart. Motion is started by displacing string from $y = a \sin \frac{\pi x}{l}$, from which it is released at $t = 0$. Show that displacement at any pt. is given by $y(x,t) = a \sin \frac{\pi x}{l} \cos \frac{\pi c t}{l}$.

$\rightarrow y(x,0) = a \sin \frac{\pi x}{l}$

$$a_n = \frac{2}{l}\int_0^l y(x,0) \sin\frac{n\pi x}{l}\, dx = \frac{2}{l}\int_0^l a \sin\frac{\pi}{l}x \sin\frac{n\pi}{l}x\, dx$$

$$= \frac{\cancel{2}a}{l}\int_0^l \left(\cos\left(\frac{n\pi}{l}-\frac{\pi}{l}\right)x - \cos\left(\frac{n\pi}{l}+\frac{\pi}{l}\right)x\right) dx$$

$$= \frac{a}{l}\left[\left[\frac{l}{(n-1)\pi}\sin\left(\frac{(n-1)\pi}{l}\right)x\right]_0^l - \left[\frac{l}{(n+1)\pi}\sin\left(\frac{(n+1)\pi}{l}\right)x\right]_0^l\right]$$

$\boxed{a_n = 0}$, $n \neq 1$

$$a_1 = \frac{2}{l}\int_0^l a \sin\frac{\pi}{l}x \sin\frac{\pi}{l}x\, dx = \frac{\cancel{2}a}{l}\int_0^l \frac{1-\cos\frac{2\pi x}{l}}{\cancel{2}}\, dx$$

$$\rightarrow a_1 = \frac{a}{l}\left[x - \frac{l}{2\pi}\sin\frac{2\pi x}{l}\right]_0^l \Rightarrow \boxed{a_1 = a}$$

Q- Solve boundary value problem $\frac{\delta^2 y}{\delta t^2} = 4\frac{\delta^2 y}{\delta x^2}$ where $y(0,t)=0$, $y(5,t)=0$, $y(x,0)=0$, $\left(\frac{\delta y}{\delta t}\right)_t = 5\sin \pi x$.

☆ 1-D Heat flow :

$$\rightarrow \quad \frac{\delta u}{\delta t} = c^2 \frac{\delta^2 u}{\delta x^2} \quad , \left\{ c = \frac{K}{S \rho} \right\}$$

conductivity

density

specific heat

$$u = X\,T$$

$$\rightarrow \quad X T' = c^2 X'' T$$

$$\rightarrow \quad \left[\cancel{\frac{X}{X''} = \frac{c^2 T}{T'} = k}\right] \left[\frac{X''}{X} c^2 = \frac{T'}{T} \right] = k$$

$$\frac{X''}{X} c^2 = k \qquad\qquad \frac{T'}{T} = k$$

$$\frac{1}{T}\, dT = k\, dt$$

$$\rightarrow \quad m^2 c^2 = k \qquad\qquad \rightarrow \quad \ln T = kt + c$$

$$\rightarrow \quad \left[m = \pm \frac{\sqrt{k}}{c} \right] \qquad\qquad \boxed{T = c_3 e^{kt}}$$

1) $\underline{k > 0}$

$$\boxed{X = C_1 e^{\frac{\sqrt{k}}{c} x} + C_2 e^{-\frac{\sqrt{k}}{c} x}}$$

$$\left\{ u = X T = e^{kt} \left[A_n e^{\frac{\sqrt{k}}{c} x} + B_n e^{-\frac{\sqrt{k}}{c} x} \right] \right\}$$

2) $\underline{k < 0}$

$$\boxed{X = C_1 \cos \frac{\sqrt{k}}{c} x + C_2 \sin \frac{\sqrt{k}}{c} x} \quad \rightarrow \quad \underline{\text{best solution}}$$

$$\boxed{u = X T = e^{kt} \left[a_n \cos \frac{\sqrt{k}}{c} x + b_n \sin \frac{\sqrt{k}}{c} x \right]}$$

3) $\underline{k = 0}$

$$\boxed{u = (C_1 x + C_2)\, C_3}$$

Q- A rod of length 'l' with insulated side initially at temp. u_0. If ends are suddenly cooled to 0°C & kept at that temp., find temp. f^n $u(x,t)$.

→ $$\frac{\delta u}{\delta t} = c^2 \frac{\delta^2 u}{\delta x^2}$$

$$\left[u(x,t) = (a_n \cos \sqrt{k}\, x + b_n \sin \sqrt{k}\, x)\, e^{-kc^2 t}\right] \left\{ \text{if solved as } \frac{X''}{X} = \frac{1}{c^2}\frac{T'}{T} \right\}$$

$u(x,0) = u_0$, $u(l,t) = 0$, $u(0,t) = 0$

$x = 0$,

$$0 = a_n e^{-kc^2 t}$$

$$\rightarrow \boxed{a_n = 0}$$

$x = l$

$$0 = b_n \sin \sqrt{k}\, l\, e^{-kc^2 t}$$

$$\rightarrow \sqrt{k}\, l = n\pi$$

$$\rightarrow \boxed{\sqrt{k} = \frac{n\pi}{l}}$$

$t = 0$

$$\cancel{b_n \sin \sqrt{k}\, x = u_0}$$

$$\cancel{\sin \sqrt{k}\, x = \frac{u_0}{b_n}}$$

$$\Rightarrow u(x,t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi}{l} x\, e^{-kc^2 t}$$

$$\rightarrow \left[u_0 = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l}\right] \rightarrow \text{half-wave sine}$$

$$\Rightarrow b_n = \frac{2}{l}\int_0^l u_0 \sin \frac{n\pi}{l} x\, dx$$

$$\rightarrow b_n = \frac{2u_0}{l}\left[-\cos \frac{n\pi x}{l}\right]_0^l \left[\frac{l}{n\pi}\right] = \frac{2u_0}{n\pi}\left[1-(-1)^n\right]$$

$$\Rightarrow \boxed{b_n = \frac{2u_0}{n\pi}\left[1-(-1)^n\right]}$$

$$\Rightarrow \boxed{u(x,t) = \sum_{n=1}^{\infty}\left(\frac{2u_0}{n\pi}\left[1-(-1)^n\right] \sin \frac{n\pi}{l} x\right) e^{-kc^2 t}}$$

★ Laplace eqⁿ for 2-D heat flow :

→ $$\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$$

$$u = XY$$

→ $$X''Y + XY'' = 0$$

→ $$\left[\frac{X''}{X} = -\frac{Y''}{Y} = k\right]$$

$$\frac{X''}{X} = k$$

$$(D^2 - k)X = 0$$

$$\boxed{D = \pm\sqrt{k}}$$

$$\frac{-Y''}{Y} = k$$

$$\boxed{m = \pm\sqrt{-k} = \pm\sqrt{k}\,i}$$

I) $k > 0$:

$$X = C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x}, \qquad Y = C_3 \cos\sqrt{k}y + C_4 \sin\sqrt{k}y$$

$$\Rightarrow \boxed{u = (C_1 e^{\sqrt{k}x} + C_2 e^{-\sqrt{k}x})(C_3 \cos\sqrt{k}y + C_4 \sin\sqrt{k}y)}$$

II) $k < 0$:

$$X = C_1 \cos\sqrt{k}x + C_2 \sin\sqrt{k}x, \qquad Y = C_3 e^{\sqrt{k}y} + C_4 e^{-\sqrt{k}y}$$

$$\Rightarrow \boxed{u = (C_1 \cos\sqrt{k}x + C_2 \sin\sqrt{k}x)(C_3 e^{\sqrt{k}y} + C_4 e^{-\sqrt{k}y})}$$

III $k = 0$:

$$X = C_1 x + C_2 \qquad Y = C_3 y + C_4$$

$$\Rightarrow \boxed{u = (C_1 x + C_2)(C_3 y + C_4)}$$

Q- Solve: $\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$, (a) $u(0,y) = \sin y$, (b) $u \to 0$ as $x \to 0$

Q- Solve $\frac{\delta^2 u}{\delta x^2} + \frac{\delta^2 u}{\delta y^2} = 0$: (i) $u \to 0$ as $y \to \infty$, (ii) $u = 0$ at $x = 0$ $\forall y$,

(iii) $u = 0$ at $x = l$ $\forall y$, (iv) $u = lx - x^2$ if

$y = 0$ $\forall x \in (0, l)$